

Universidad Autónoma de Nuevo León  
Facultad de Ingeniería Mecánica y Eléctrica

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Maestría en Ciencias de la Ingeniería Eléctrica  
Unidad de Aprendizaje: **Control No Lineal**

## Proyecto Integrador de Aprendizaje

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Estudiante: Roberto Treviño Cervantes  
Matrícula: 1915003  
Maestro: Dr. Alberto Cavazos

## Case study

*Yield-Aware Higher-Order Trajectory Optimization for Step-and-Scan Wafer Stages.*

The commercial performance of a scanner is ultimately measured in wafers per hour. The time spent stepping and *settling* — the interval between the end of a step motion and the moment the stage has damped its residual vibrations to within the nanometre-level positioning specification — represents a significant and controllable fraction of the total cycle time, which places the allowable settle time below 10 ms. The key lever for reducing settling time is the generation of *reference trajectories*: motion profiles that command the stage from its current position to the next scanning start-point with minimal excitation of mechanical resonances.

## Model

Al-Rawashdeh et al. propose modeling the lithography scanner as a combination of three open-loop serial chains: the reticle chain ( $j = 1$ ), the optics chain ( $j = 2$ ), and the wafer chain ( $j = 3$ ). A dual-stage architecture is employed: a long-stroke motor (LS) for large-range travel of at least 300 mm with micrometer accuracy, and a short-stroke motor (SS) with Lorentz actuators for nanometric, contact-free positioning. The physical connections between stages are represented by lumped-parameter spring-damper systems according to the Kelvin-Voigt model.

The full 6-DOF model is restricted to in-plane motions, since rotations about  $x$ ,  $y$  and translation on  $z$  are used mainly for focusing. The machine comprises 16 bodies; the global state vector has  $6 \times 16 = 96$  states.

## Methodology

Trajectories that have smooth higher-order derivatives result in less excitation of poorly damped modes in the structure. We implemented a simulation of the multi-body model using the MuJoCo physics engine for Python. The simulation serves as a virtual test bed where the proposed trajectories can be validated under the dynamic model.

The Kelvin-Voigt couplings are implemented as MuJoCo joints with explicit **stiffness** and **damping** parameters. For the actuated stages, MuJoCo position actuators are used with  $K_P = 10^7$  at  $dt = 10^{-4}$  s. During the scanning phase, the wafer stage moves at  $v_{w,\text{scan}} = 0.8$  m/s while the reticle moves in the opposite direction at  $v_{r,\text{scan}} = 3.2$  m/s, maintaining the demagnification ratio  $\alpha = 0.25$ .

## CNN cost function and validation

The contribution of this work is the forward direction of the defect-prediction problem: given trajectory parameters  $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$ , predict the expected wafer defect pattern by means of a CNN trained on the WM-811K dataset of real wafer maps. The cost function penalizes the cumulative probability mass on the non-plausible defect classes:

$$J(\mathbf{r}) = \sum_{c \notin \mathcal{C}_{\text{scan}}} \hat{p}_c(\mathbf{r}), \quad \mathcal{C}_{\text{scan}} = \{\text{none}, \text{scratch}, \text{edge-local}, \text{random}\}. \quad (1)$$

The methodology is valid when synthetic wafer maps activate only the classes in  $\mathcal{C}_{\text{scan}}$  and  $J(\boldsymbol{\pi})$  shifts reproducibly under independent sweeps of  $(v, a, j, s)_{\max}$ . The CNN must never activate *donut* or *near-full*, which correspond to etching/cleaning patterns rather than trajectory-induced defects.

## Resultados

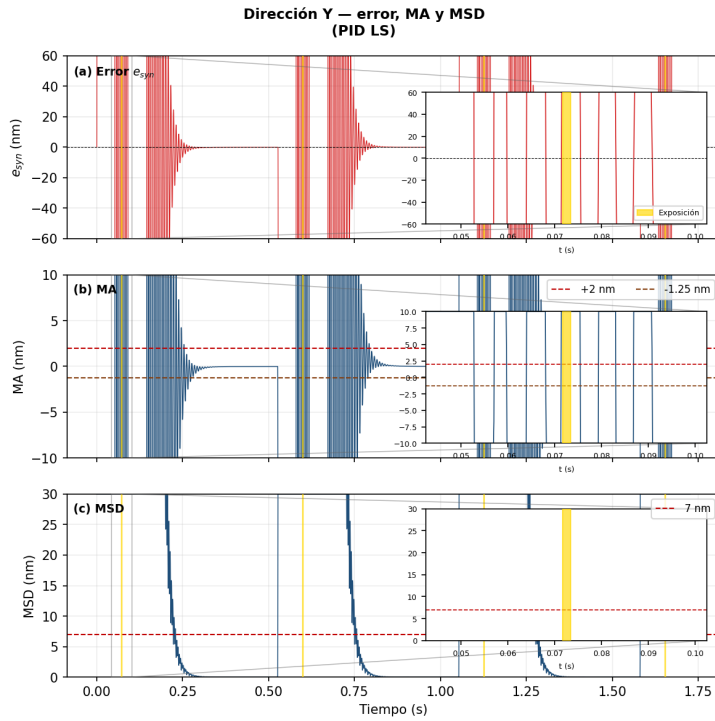
Control PID + *feedforward*.

$$u(t) = \underbrace{K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}}_{\text{PID}} + \underbrace{m \ddot{r}(t)}_{\text{FF}}$$

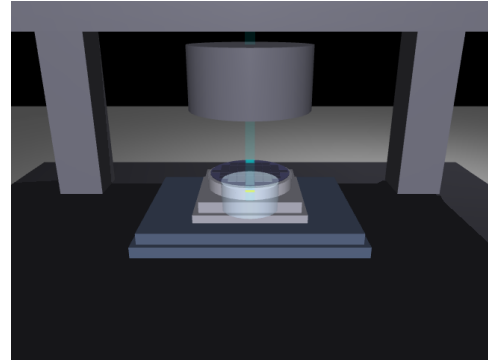
$e(t) = r(t) - y_{\text{larga}}(t)$  (realimentación sobre la etapa larga).

Ancho de banda  $f_{bw} = 50$  Hz,  $\omega_c = 2\pi f_{bw}$ ,  $\alpha \approx 3$ .

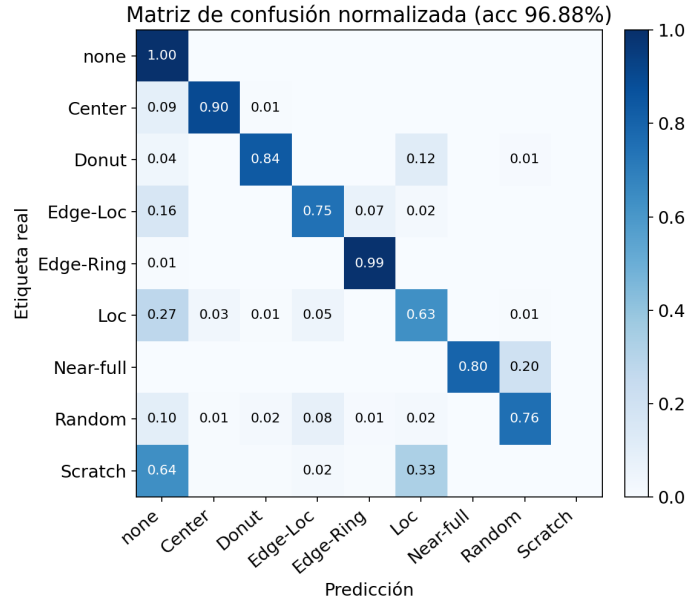
$$\begin{aligned} K_p &\approx 8.66 \times 10^6 \text{ N/m} & K_p &= m (2\pi f_{bw})^2 \\ K_i &\approx 2.72 \times 10^8 \text{ N/(m}\cdot\text{s)} & K_d &= m \omega_c \\ K_d &\approx 2.75 \times 10^4 \text{ N}\cdot\text{s/m} & K_i &= K_p \omega_c / \alpha^2 \\ m &\approx 87 \text{ kg} \end{aligned}$$



Error  $e_{syn}$ , MA y MSD (dir. Y).



Entorno de simulación MuJoCo.



Matriz de confusión — 3 conv + 2 FC, 96.88 % accuracy.

## Conclusion

The forward direction explored here — trajectory parameters → predicted defect pattern — complements the established industrial inverse problem of inferring process causes from observed wafer defect maps. The CNN-based cost function enables trajectory optimization to proceed entirely in simulation, without access to physical lithographic equipment, and provides a reproducible link between the kinematic parameters of the reference trajectory and the predicted yield signature on the wafer.